Concise Manual for `compare_urdf_dh_frames.m`

Robot System Identification utilities

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1 Purpose

`compare_urdf_dh_frames.m` checks a standard Denavit–Hartenberg (DH) model generated from a URDF serial chain. At a user-supplied joint configuration, it reconstructs the DH frames, evaluates the same configuration with the URDF model, and compares:

- every URDF joint axis with the corresponding DH axis z_{i-1}; and
- the URDF and DH reconstructions of the selected tool pose.

It also creates a figure containing the URDF model, the DH chain, and an aligned overlay. The comparison is diagnostic and does not modify its inputs.

2 Requirements and inputs

The function requires MATLAB with Robotics System Toolbox. Add the utility folder to the MATLAB path before calling it. Its three file inputs are:

URDF file The source robot description.

DH CSV The generated `dh_table.csv`, including the columns `joint`, `a_m`, `alpha_rad`, `d_m`, `theta_rad`, `t`, and `s`.

Kinematics JSON The generated `urdf_kinematics.json`, containing the base-to-first-DH-frame and last-DH-frame-to-tool transforms.

These generated files are documented in `urdf_to_standard_dh_manual.pdf`.

The vector `q` must contain one finite joint coordinate per CSV row, in the same order as the CSV. Angles are expressed in radians and prismatic coordinates in metres.

3 Function interface

```
result = compare_urdf_dh_frames( ...
    urdfPath, dhCsvPath, kinematicsJsonPath, q, ...
    'BaseLink', baseLink, ...
    'ToolLink', toolLink, ...
    'FrameScale', 0.08, ...
    'ShowVisuals', true);
```

`BaseLink` and `ToolLink` are required. They select the URDF chain to compare and prevent unrelated branches or fixed frames from entering the test. `FrameScale` is the plotted coordinate-axis length in metres; its default is 0.08. `ShowVisuals` enables URDF visual meshes and is `true` by default. Setting it to `false` can make complex models faster and clearer to inspect.

4 Example

```
addpath(fullfile('src', 'utils', 'UrdfToDH'));

robotRoot = fullfile('src', 'Identification_Luna_DB3.5', ...
    'RobotDescription');
q = zeros(1, 7);

result = compare_urdf_dh_frames( ...
    fullfile(robotRoot, 'urdf', 'arm_with_None.urdf'), ...
    fullfile(robotRoot, 'DH', 'dh_table.csv'), ...
    fullfile(robotRoot, 'DH', 'urdf_kinematics.json'), ...
    q, ...
    'BaseLink', 'starman_arm_link_cart', ...
    'ToolLink', 'starman_arm_ids');
```

Run the comparison at several representative configurations, not only at the zero pose. Configurations must respect the URDF joint-coordinate convention used to generate the DH model.

5 Computation and displayed result

For row i, the function evaluates the standard-DH transform

$$A_i = \mathrm{RotZ}(\theta_i^*)\,\mathrm{TransZ}(d_i^*)\,\mathrm{TransX}(a_i)\,\mathrm{RotX}(\alpha_i), \tag{1}$$

where

$$\theta_i^* = \theta_i + (1 - t_i) s_i q_i, \qquad d_i^* = d_i + t_i s_i q_i. \tag{2}$$

The residual transforms from the JSON file align the reconstructed DH chain with the selected URDF base and tool.

The figure has three panels: URDF model and body frames, generated DH frames, and their overlay. DH frames are labelled `D0`, `D1`, and so on; URDF joint axes are labelled `UJ1`, `UJ2`, and so on. The command window reports the tool position and orientation errors and, for each joint, the axis-angle error and shortest distance between the URDF and DH axis lines.

6 Returned structure and interpretation

The returned `result` structure contains joint names and configuration, the URDF and DH tool transforms, every reconstructed DH frame, joint-axis angle and line-distance errors, tool position and orientation errors, and the figure handle.

For a consistent conversion, all reported errors should be close to numerical zero, subject to the precision of the URDF and generated files. A large tool error indicates a kinematic mismatch. A large per-joint angle error indicates an axis-direction mismatch, while a large line distance indicates that the two compared axes are displaced.

Under the standard-DH convention, URDF joint i is associated with z_{i-1}, not z_i. URDF body frames and DH frames are defined for different purposes and therefore do not need to coincide visually; joint-axis and final tool-pose agreement are the relevant checks.